

A PCA Study of Bike Rentals and Temperature

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ABSTRACT

The goal of this project is to use real daily bike sharing data from Washington, D.C. to answer a clear question: do people rent more bikes when the temperature is higher? The dataset comes from the Capital Bikeshare `day` file (2011 to 2012), where each row represents one day. The main variables used are daily temperature and total daily rentals. Before analysis, the data are cleaned and prepared so the results are consistent and reproducible, including checking for missing values, selecting the final columns, and making sure units and scaling are handled correctly. To capture the main joint pattern between temperature and rentals, the project uses PCA. The analysis compares covariance and correlation based views, and explains why the normalization choice $1/(n-1)$ versus $1/n$ matters for the sample covariance. PCA outputs like eigenvalues, percent variance explained, and loading vectors are used to interpret the dominant direction of variation. Score plots and related visual checks are then used to see whether higher temperature tends to align with higher bike usage in the main component structure.

Keywords:

1 PROBLEM FORMULATION AND DATA PROVENANCE

Research Question

This project focuses on one simple question:

Do people rent more bikes when temperature is higher?

To study this, PCA is used to study the main shared pattern between daily temperature and daily bike rentals.

Chosen Dataset

The data comes from the public Bike Sharing dataset (Capital Bikeshare, Washington D.C.) from the UCI Machine Learning Repository. The daily file `day.csv` has one row per day for 2011 and 2012, so there are

$$n = 731$$

days total.

Two columns are used:

- `temp`: normalised daily average temperature,
- `cnt`: total daily rentals (casual + registered).

Variables and units

For day i :

$$T_i = 41 \text{ temp}_i \quad \text{and} \quad C_i = \text{cnt}_i.$$

So the raw data for each day is the pair

$$R_i = (T_i, C_i),$$

and stacking all days gives

$$R = \begin{bmatrix} T_1 & C_1 \\ \vdots & \vdots \\ T_n & C_n \end{bmatrix} \in \mathbb{R}^{n \times 2}.$$

Preprocessing steps

Before running PCA, the data are centered and scaled so the two variables are comparable.

1. Centering

$$\mu_T = \frac{1}{n} \sum_{i=1}^n T_i, \quad \mu_C = \frac{1}{n} \sum_{i=1}^n C_i,$$

$$T_i^c = T_i - \mu_T, \quad C_i^c = C_i - \mu_C.$$

2. Scaling Using the sample standard deviations:

$$\sigma_T^2 = \frac{1}{n-1} \sum_{i=1}^n (T_i - \mu_T)^2, \quad \sigma_C^2 = \frac{1}{n-1} \sum_{i=1}^n (C_i - \mu_C)^2,$$

the standardised (z score) values are:

$$Z_{i1} = \frac{T_i - \mu_T}{\sigma_T}, \quad Z_{i2} = \frac{C_i - \mu_C}{\sigma_C}.$$

The final matrix used for PCA is

$$X = \begin{bmatrix} Z_{11} & Z_{12} \\ \vdots & \vdots \\ Z_{n1} & Z_{n2} \end{bmatrix} \in \mathbb{R}^{n \times 2}.$$

Centering, as we divide by the standard deviation, makes everything measured relative to an “average day”, so the PCA focuses on day to day changes.

Scaling matters because rentals are in the thousands while temperature is in the tens. Without scaling, PCA would mostly follow the rentals column just because the numbers are bigger. With z scores, both columns are on the same scale (each is measured in standard deviations), and PCA is effectively working on the correlation between temperature and rentals.

An important point to note is that X_i is being treated as a row vector (one day, p features). With that, the convention changes, and is defined as $(X_i - \bar{X})^\top (X_i - \bar{X})$ is a $p \times p$. This is an outer product, so summing it produces a covariance matrix across features, not a scalar. Hence, our setup concludes with what is shown in the project document:

Let

$$X \in \mathbb{R}^{n \times p}$$

be the column-centered data matrix, with rows equal to observations and columns equal to features. The i -th row is written as $X_i \in \mathbb{R}^p$. The sample mean vector is

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Two versions of the sample covariance matrix are considered:

$$S_{\text{unb}} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^\top (X_i - \bar{X}), \quad S_{\text{mle}} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^\top (X_i - \bar{X}).$$

The first uses $1/(n-1)$, the second uses $1/n$. In the bike dataset one has $n = 731$ and $p = 2$.

2 ANALYTICAL WRITE-UP (BIAS, CONSISTENCY, SUFFICIENCY, EFFICIENCY)

2.1 Bias

Derivation 1: Shown is unbiasedness of S_{unb}

Assume the rows are independent and identically distributed with

$$\mathbb{E}[X_i] = \mu, \quad \text{Cov}(X_i) = \Sigma.$$

Define

$$Y_i = X_i - \mu, \quad \bar{Y} = \bar{X} - \mu.$$

Then

$$X_i - \bar{X} = Y_i - \bar{Y}.$$

The matrix inside both covariance estimators is

$$\sum_{i=1}^n (X_i - \bar{X})^\top (X_i - \bar{X}) = \sum_{i=1}^n (Y_i - \bar{Y})^\top (Y_i - \bar{Y}).$$

Using the standard identity

$$\sum_{i=1}^n (Y_i - \bar{Y})^\top (Y_i - \bar{Y}) = \sum_{i=1}^n Y_i^\top Y_i - n \bar{Y}^\top \bar{Y},$$

gives

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - \bar{X})^\top (X_i - \bar{X}) \right] = \mathbb{E} \left[\sum_{i=1}^n Y_i^\top Y_i \right] - \mathbb{E} [n \bar{Y}^\top \bar{Y}].$$

Now

$$\mathbb{E}[Y_i^\top Y_i] = \Sigma \quad \because \quad \text{Cov}(Y_i) = \Sigma,$$

so

$$\mathbb{E} \left[\sum_{i=1}^n Y_i^\top Y_i \right] = n\Sigma.$$

Now, because

$$\text{Cov}(\bar{Y}) = \frac{1}{n} \Sigma \quad \because \quad \bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \text{ and the rows are i.i.d.,}$$

hence

$$\mathbb{E}[\bar{Y}^\top \bar{Y}] = \frac{1}{n} \Sigma \Rightarrow \mathbb{E}[n \bar{Y}^\top \bar{Y}] = \Sigma.$$

Giving us

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - \bar{X})^\top (X_i - \bar{X}) \right] = n \Sigma - \Sigma = (n-1) \Sigma \quad \therefore \quad \mathbb{E}[S_{\text{unb}}] = \Sigma.$$

Therefore, S_{unb} is an unbiased estimator of the population covariance matrix.

Derivation 2: Shown is bias of S_{mle} and Bessel correction

By definition,

$$S_{\text{mle}} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^\top (X_i - \bar{X}) = \frac{n-1}{n} S_{\text{unb}}.$$

Taking expectations,

$$\mathbb{E}[S_{\text{mle}}] = \frac{n-1}{n} \mathbb{E}[S_{\text{unb}}] = \frac{n-1}{n} \Sigma \quad \therefore \quad \mathbb{E}[S_{\text{unb}}] = \Sigma.$$

Therefore S_{mle} is biased:

$$\mathbb{E}[S_{\text{mle}}] = \left(1 - \frac{1}{n}\right) \Sigma.$$

Multiplying S_{mle} by the factor

$$\frac{n}{n-1}$$

removes this bias and gives us S_{unb} . This simple scaling is Bessel's correction.

For the bike dataset,

$$\frac{n-1}{n} = \frac{730}{731} \approx 0.9986,$$

so the bias is theoretically important but numerically very small.

Effect of bessel correction on PCA and proportions of variance

Let (λ_j, v_j) be an eigenpair of S_{unb} , so

$$S_{\text{unb}} v_j = \lambda_j v_j.$$

Since

$$S_{\text{mle}} = \frac{n-1}{n} S_{\text{unb}},$$

giving us

$$S_{\text{mle}} v_j = \frac{n-1}{n} S_{\text{unb}} v_j = \frac{n-1}{n} \lambda_j v_j \quad \therefore \quad \lambda_j^{(\text{mle})} = \frac{n-1}{n} \lambda_j^{(\text{unb})}.$$

Bessel's correction is the adjustment that replaces the $1/n$ scaling in a sample variance or covariance with $1/(n-1)$. (1) This is done so the mean is estimated from the same sample so the raw average of squared deviations tends to come out much smaller. (1) In this context that means the MLE covariance S_{mle} underestimates the population covariance by a uniform factor of $(n-1)/n$, and the unbiased covariance

S_{unb} corrects for it by multiplying the same scatter matrix by $n/(n-1)$.⁽¹⁾ That is why changing from S_{mle} to S_{unb} rescales all PCA eigenvalues by approximately $n/(n-1)$, while changing the other way rescales them by $(n-1)/n$.⁽¹⁾

Scaling has effects on the PCA. This is because eigenvalues measure variance along each principal direction, so multiplying the covariance matrix by a constant multiplies every eigenvalue by that same constant. However, the *proportions* of variance are ratios of one eigenvalue to the sum of all eigenvalues. Since both the numerator and denominator get multiplied by the same constant, the constant cancels. Therefore, the PVE variance percentages do not change at all when using S_{mle} or S_{unb} . The principal directions are identical, the ordering of components is identical, and the proportions are identical.

On this dataset, $n = 731$, so the correction factor $n/(n-1)$ is extremely close to 1. Numerically, the eigenvalues from S_{unb} are only about 0.14 percent larger than those from S_{mle} . However, it is still worth stating that the S_{unb} matches the population covariance in expectation, and the S_{mle} is uniformly shrunk because the mean was estimated from the same data.

2.2 Consistency

Consistency of the sample mean $\bar{X}$

Assume the rows $X_1, \dots, X_n \in \mathbb{R}^P$ are i.i.d. with finite second moments and

$$\mathbb{E}[X_i] = \mu.$$

The sample mean is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

By the law of large numbers (it is applied for every coordinate),

$$\bar{X}_n \xrightarrow{P} \mu \quad \therefore \quad \bar{X}_n \text{ is a consistent estimator of } \mu.$$

Consistency of the unbiased covariance S_{unb}

Let $\Sigma = \text{Cov}(X_i)$. The unbiased sample covariance is

$$S_{\text{unb}}^{(n)} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^\top (X_i - \bar{X}_n).$$

Using the standard expansion,

$$S_{\text{unb}}^{(n)} = \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n X_i^\top X_i - \bar{X}_n^\top \bar{X}_n \right).$$

Now,

$$\frac{1}{n} \sum_{i=1}^n X_i^\top X_i \xrightarrow{P} \mathbb{E}[X_1^\top X_1] \quad \therefore \quad \text{law of large numbers,}$$

and

$$\bar{X}_n^\top \bar{X}_n \xrightarrow{P} \mu^\top \mu \quad \therefore \quad \bar{X}_n \xrightarrow{P} \mu \text{ and products are continuous.}$$

Also $\frac{n}{n-1} \rightarrow 1$. Therefore,

$$S_{\text{unb}}^{(n)} \xrightarrow{P} \Sigma \quad \therefore \quad S_{\text{unb}}^{(n)} \text{ is a consistent estimator of } \Sigma.$$

Using the law of large numbers, the sample averages in this dataset become more stable as more days are included. The sample mean is just an average across days, identically distributed observations with

finite variance, it converges to the true population mean. The same idea also applies to the second order quantities used in covariance, since the sample averages of products like $(X - \bar{X})^\top (X - \bar{X})$ settle toward their true population values as n grows. The unbiased covariance S_{unb} is built entirely from these parts, except it centres using the sample mean instead of the true mean. That substitution does not change the limit because the sample mean itself converges, so the centring becomes correct in the large-sample sense. After that, the continuous mapping idea applies, because each covariance entry is a continuous function of the empirical mean and the empirical second moments, so once those inputs converge, the covariance matrix converges as well. (2) Therefore, $\bar{X}$ and S_{unb} are consistent estimators of μ and Σ . This is exactly why the PCA results stop being noisy once enough days are included.

Resampling experiment

A simple increasing-sample experiment is used to empirically demonstrate consistency. The data are ordered by time, and for sample sizes $m = 30, 40, \dots, n$ the first m days are used to compute:

- (i) the two components of the sample mean of the standardized matrix X
- (ii) the diagonal entries of the unbiased covariance S_{unb}
- (iii) the leading PCA eigenvalue

The resulting paths are plotted in Figure 1.

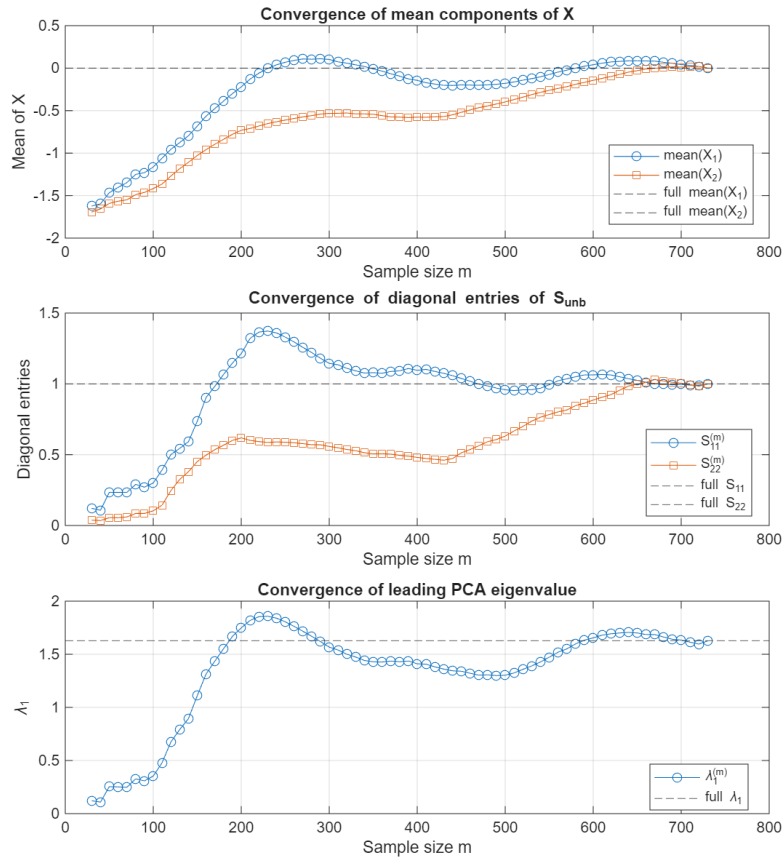


Figure 1. Convergence of the sample mean, diagonal entries of S_{unb} , and the leading PCA eigenvalue as more days are added.

The graphs show a clear pattern where estimates are unstable when only a small number of days are used, and then become much more stable as more data are added. In the first graph, the two components of the sample mean change significantly at small sample sizes, but gradually level off and approach the full-sample reference values. This reflects the fact that averages based on only a few days are strongly

influenced by short-term weather and demand conditions. In the second graph, the diagonal entries of S_{unb} start far from their long-run values and then move steadily toward one, which is expected for standardized features once enough observations are included. The third graph shows similar behavior for the leading eigenvalue. It increases quickly at first, fluctuates in the early part of the sample, and then settles near the full-sample value. The plots very clearly indicate that once the sample size reaches a few hundred days, the estimates become much less sensitive to individual observations, and the main PCA structure is no longer influenced by early data points.

2.3 Sufficiency

Derivation 3: Shown is sufficiency for the mean in the univariate normal case

Let $X_1, \dots, X_n$ be i.i.d. from $\mathcal{N}(\mu, \sigma^2)$ with σ^2 treated as known. The joint density is

$$f(x_1, \dots, x_n; \mu) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

Use the identity

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2.$$

Substituting into the likelihood gives

$$f(x_1, \dots, x_n; \mu) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \bar{x})^2\right) \cdot \exp\left(-\frac{n}{2\sigma^2} (\bar{x} - \mu)^2\right).$$

This is a factorization of the form

$$f(x_1, \dots, x_n; \mu) = h(x_1, \dots, x_n) \phi(\bar{x}; \mu),$$

where h does not depend on μ , and ϕ depends on the data only through $\bar{x}$. Therefore, by the factorization theorem, $\bar{X}$ is a sufficient statistic for μ .

Sufficiency means that a statistic captures all the information in the data about a parameter, so once that statistic is known, the remaining observations add nothing further for inference.(4) This idea matters because it gives a precise meaning to the claim that no relevant parameter information is lost under a model assumption.(3) In the univariate normal case with known variance, fixing the sample mean leaves only the term $\sum (x_i - \bar{x})^2$, which does not involve μ , so it cannot improve inference about the mean.(3) In the multivariate normal setting with unknown (μ, Σ) , the same reasoning applies at a larger scale. The likelihood can be written so that $\bar{X}$ contains the information about location, while S contains the information about spread and how variables move together, and together they preserve all information in the sample about the model parameters.(5) Since PCA is built from the covariance structure, it uses S as its input, and under multivariate normality this choice is consistent with the sufficiency result for Σ .(5)

Under the multivariate normal model, the sample mean $\bar{X}$ and sample covariance S together form a sufficient statistic for the unknown parameters (μ, Σ) .(5) PCA is driven entirely by the covariance structure. The principal directions are the eigenvectors of S , and the explained variances are its eigenvalues. Therefore PCA is a deterministic function of S alone. In this Gaussian setting, working with S does not remove covariance information that would matter for inference about Σ .

2.4 Efficiency

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known. Full derivation not shown as two previous derivations done. The likelihood is

$$\ell(\mu) = C - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Differentiate:

$$\ell'(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu), \quad \ell''(\mu) = -\frac{n}{\sigma^2}.$$

Thus the Fisher information is

$$I(\mu) = -\mathbb{E}[\ell''(\mu)] = \frac{n}{\sigma^2}.$$

CRLB and efficiency of $\bar{X}$

The Cramér–Rao lower bound for any unbiased estimator T of μ is

$$\text{Var}(T) \geq \frac{1}{I(\mu)} = \frac{\sigma^2}{n}.$$

The sample mean $\bar{X}$ is unbiased and

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Full derivation is not shown as two previous derivations are done.

Efficiency here simply means:

If the data really are i.i.d. normal with known variance, then no unbiased method can estimate μ with less spread than σ^2/n . The Cramér–Rao bound gives that lowest possible variance, and the sample mean $\bar{X}$ actually reaches it. So $\bar{X}$ is not only unbiased, it is also as precise as possible for this model. As n grows, the uncertainty reduce similar to $1/n$, which matches the idea that averaging more independent measurements gives a more stable estimate of the true mean.

comparison between the sample mean and sample median

If the variance is unknown, the setup changes because there are now two unknown parameters instead of one.(6) Even so, the sample mean still performs very well. In the normal model, the maximum likelihood estimate of μ is still $\bar{X}$, and the maximum likelihood estimate of σ^2 is based on the average squared deviation from $\bar{X}$.(6) When n is large, maximum likelihood estimators in standard models are known to behave nearly as well as possible, in the sense that their variance approaches the best theoretical limit as the sample size grows.(6) For this reason, even with unknown variance, $\bar{X}$ remains the default estimator for μ and is essentially the best choice under normality for large samples.(6)

For the mean versus the median under normality, the short answer is that both target the same center, and both are unbiased for μ in a symmetric normal distribution, but the mean is more efficient.(7) The mean uses the full numeric values, while the median depends mainly on the ranking of observations.(7) Under truly normal data, the median has a wider sampling spread than the mean, so it generally needs more observations to reach the same accuracy.(7) The main advantage of the median is robustness. It is less affected by outliers, so if the data is not well described by a normal distribution or contain extreme values, the median can be more reliable in practice even though it is less efficient under the ideal normal model.(7)

2.5 Empirical distributions & convergence

For a single feature $X_1, \dots, X_n$, the empirical CDF is

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{X_i \leq x\}.$$

It is built directly from the data. For any value x , $\hat{F}_n(x)$ is simply the fraction of observations that are at or below x .(8)

The KS test compares the empirical CDF from the data to a model or ideal CDF. It looks at the largest vertical gap between the two curves over all x values. If that maximum gap is large, the data do not look like it came from the model distribution. If it is small, the model distribution is a reasonable fit in this max-gap sense. The standard KS p-value is derived for a reference CDF that is fixed in advance. If the model parameters are first fitted using the same data, then the usual KS p-value is no longer exact, and a corrected method is normally used instead.(9)

Glivenko–Cantelli says that for i.i.d. data from a true distribution, the empirical CDF approaches the true CDF as the sample size increases, and this happens across the entire range of x , not just at a single point.(10) This is why KS-type checks are useful for assessing distribution assumptions. The KS idea looks at the largest gap between the empirical CDF and the model CDF anywhere on the plot. If the model is a good match, that maximum gap should shrink as more data are added. If the model is a poor match, the gap often stays visible, and the ECDF plot helps pinpoint where the model misses.(10)

In this project, the empirical CDF is a direct way to check whether the standardized bike features behave like the simple models people often assume in PCA write-ups. For example, after converting temperature to Celsius and z-scoring temperature and rental count, the ECDF of each variable can be plotted against a normal CDF to see if the bulk of the data and the tails look roughly normal. The KS test then gives one clean number, the maximum vertical gap between the ECDF and the model CDF, which makes it easy to compare model fit across variables. This is useful here because rentals are often skewed and can have heavier tails than a normal curve, especially during unusual days, and the ECDF makes that visible so accurate analysis can be carried out further. Glivenko–Cantelli explains why this method is meaningful as the dataset grows. With more days, the ECDF becomes a more reliable picture of the true distribution, so large and persistent gaps are more likely to reflect the real model mismatching and not just from sampling noise.

3 EXPERIMENTAL COMPONENT (PCA ON A LARGE DATASET)

3.1 Build the data matrix

Features and units. Two columns are used.

- `temp_C`: daily average temperature in degrees Celsius.
- `cnt`: total daily rentals in trips.

3.2 Build the data matrix

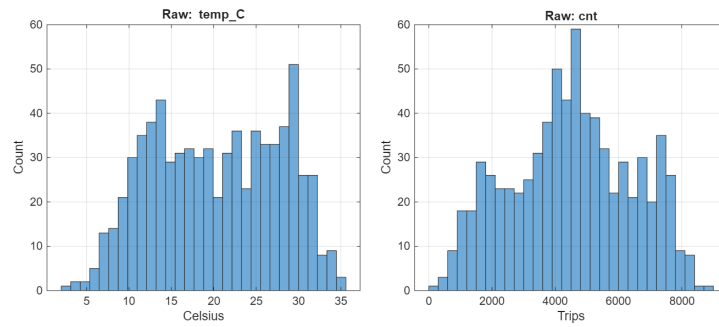


Figure 2. Raw distributions for temperature (Celsius) and total rentals (trips).

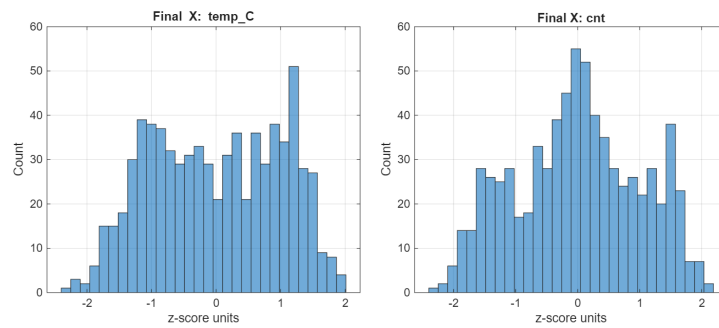


Figure 3. Distributions after centering and standardising, shown in z-score units.

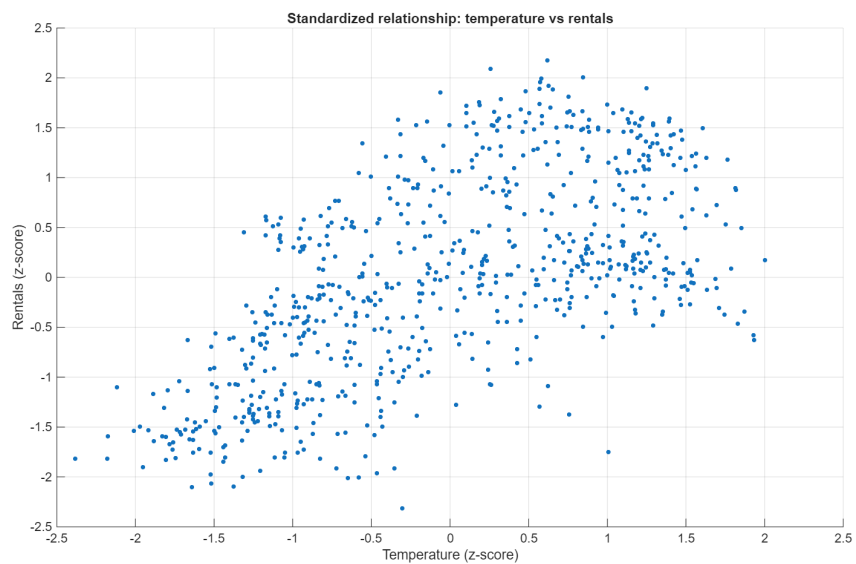


Figure 4. Standardised scatter plot of temperature and rentals in z-score units.

Table 1. Summary statistics for the two selected features (daily data).

Feature	Units	Min	Max	Mean	Std (n-1)
temp_C	Celsius	2.424	35.328	20.311	7.505
cnt	Trips	22	8714	4504.35	1937.21

Figure 2 shows the raw values before PCA. The temperature range is for a full seasonal cycle, and rentals range from very low demand days up to very high demand day. These are treated as peak days. Table 1 simply shows them for easier reading. The problem here is the scale. As temperature varies by tens of degrees, while rentals vary by thousands of trips, so a covariance-based PCA on raw units would be pointless as it would just get dominated by rentals purely due to the units.

Centering is needed because PCA is meant to describe variation around the average day. If the columns are not centered, the first component can get pulled toward the mean, and the covariance calculation is no longer describing pure spread around the center. After centering, the covariance and principal directions reflect how temperature and rentals vary together relative to their daily averages.

Scaling by the standard deviation is used here because the two features are measured in different units and different sizes. Standardising makes both columns equal, so PCA focuses on how these variables move together rather than whichever variable has larger raw numbers. Figure 3 shows that this step causes both variables to be in z-score units, where $+1$ means one standard deviation above average and -1 means one standard deviation below average.

Figure 4 shows the same scatter plot after we subtract the averages and scale both variables by their standard deviations. This is the exact form needed in the PCA input matrix. Because both axes are now on the same z-score scale, the main slope of the point cloud reflects the real pattern between temperature and rentals, not just the fact that rentals are measured in much bigger numbers than temperature. (14)

3.3 Compute PCA

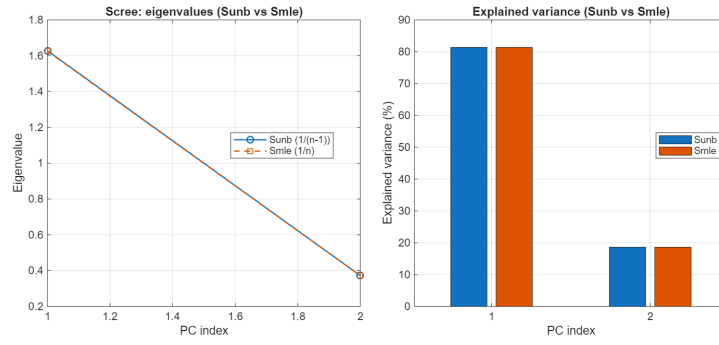


Figure 5. Eigenvalues and explained-variance comparison for S_{unb} and S_{mle} .

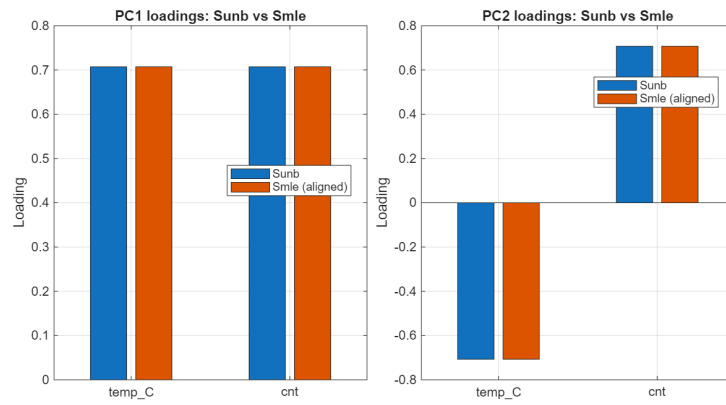


Figure 6. PC1 and PC2 loading vectors for S_{unb} and S_{mle}

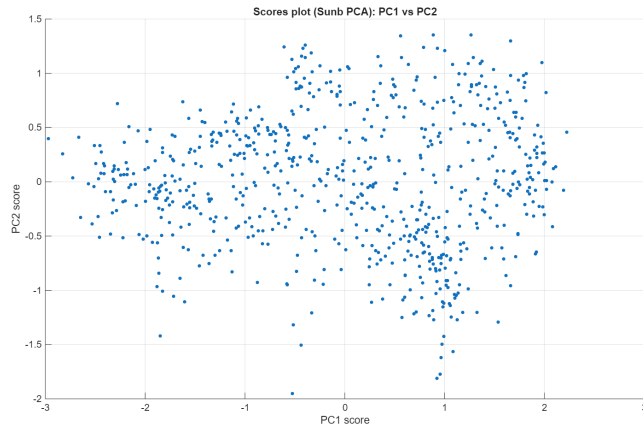


Figure 7. Scores plot for S_{unb} PCA showing PC1 vs PC2.

PC	λ_{unb}	Var. explained (%)	λ_{mle}	Var. explained (%)
1	1.6275	81.37	1.6253	81.37
2	0.3725	18.63	0.3720	18.63

Table 2. PCA eigenvalues and explained variance for S_{unb} and S_{mle} .

PC	Feature	Loading (S_{unb})	Loading (S_{mle})
1	temp_C	0.7071	0.7071
1	cnt	0.7071	0.7071
2	temp_C	-0.7071	-0.7071
2	cnt	0.7071	0.7071

Table 3. Loading vectors for PC1 and PC2 where the S_{mle} signs are aligned to match S_{unb}

The PCA directions do not change in any significant way between S_{unb} and S_{mle} . Figure 6 shows that the loading bars sit on top of each other once the sign alignment is applied, which is expected because S_{mle} is just a constant rescaling of the same scatter matrix. The only visible difference is in Figure 5, where the eigenvalues under S_{mle} are slightly smaller in absolute value, while the explained-variance percentages remain identical.

This matches the theory from the bias discussion, where multiplying a covariance matrix by a constant changes eigenvalues but does not change eigenvectors or explained-variance ratios. The scatter plot in Figure 4 is clearly stretched along an upward diagonal, so the first PCA direction points roughly from the bottom-left to the top-right, meaning temperature and rentals tend to move together.

Days with positive temperature z-scores usually sit in the positive rentals region more often than not, and days with negative temperature z-scores tend to sit in the negative rentals region, which matches the positive pattern. The points that break this pattern show up in the mixed-sign quadrants, for example warmer-than-average days with below-average rentals or cooler-than-average days with above-average rentals, and these are the kinds of deviations that the second direction picks up once the main diagonal trend is removed. In terms of whether eigenvector directions change significantly, they should not in this setup when comparing S_{unb} and S_{mle} because the difference between them is only a uniform scaling of the same matrix. So the PCA directions stay essentially the same, and any difference would only be insignificant numerical rounding. This would not have any significant outcome.(11)

observations and analysis of PC1 and PC2

The loadings in Table 3 make the analysis very simple as PC1 has positive and equal loadings on temperature and rentals, so PC1 increases when both temperature and rentals increase together. Days with warm weather and high rental counts land on the positive side of PC1, while colder and quieter days land on the negative side.

PC2 has opposite signs, so it measures the contrast between rentals and temperature. Positive PC2 means rentals are high relative to the daily temperature, and negative PC2 means rentals are low relative to the temperature. The scatter plot in Figure 7 spreads much more along PC1 than PC2, which is consistent with Table 2 where PC1 explains about 81% of the variance and PC2 explains about 19%.

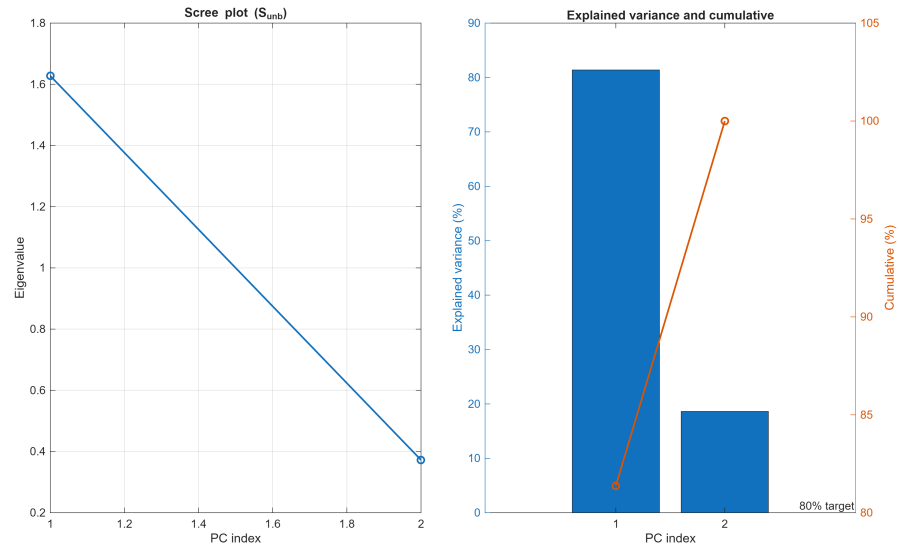


Figure 8. Scree plot and explained-variance target used to choose the reduced rank k from S_{unb} .

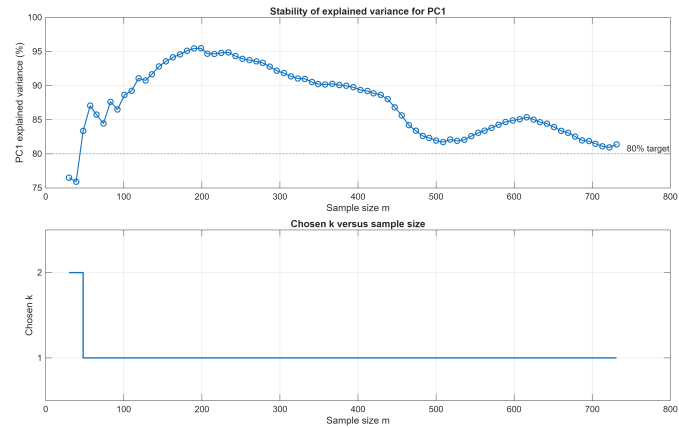


Figure 9. Stability of PC1 explained variance and the chosen k as the sample size increases.

Figure 8 supports setting $k = 1$. The scree curve drops sharply from PC1 to PC2, so most of the variance sits in the first direction, and PC2 is much smaller. Using an explained-variance target is a standard rule for selecting k , and here PC1 alone covers the 80% target, so one component already captures the main pattern in the data.(14) Since this dataset only has two variables, k can only be 1 or 2. Choosing $k = 2$ would always give 100% variance explained. For this project, the goal is the dominant joint pattern between temperature and rentals, so $k = 1$ is option that makes the most sense.(13)

Figure 9 shows why that choice is also better for stability as more days are added. Initially, the PC1 explained variance moves around a lot because a small window of days can be dominated by specific weather weeks or unusual demand. As m grows, the PC1 percentage settles and stays above the 80% line, and the chosen k quickly becomes 1 and then remains 1 for essentially the whole range. In practical terms, this means the decision to set $k = 1$ is correct. It does not depend on a lucky subset of days, and it stays consistent once the sample includes a few hundred daily records.

ETHICS & LIMITATIONS

This project uses a public dataset that is already aggregated by day, so it does not contain personal identifiers and the privacy risk is low. Still, the results should be handled carefully because the analysis is descriptive. A positive relationship between temperature and rentals does not prove temperature alone causes higher demand, since seasonality, holidays, commuting patterns, pricing, and service availability can shift rentals at the same time as weather. The dataset is also specific to one city and one system, so the findings may not transfer to places with different infrastructure, cultures, or climate. On the method side, PCA is linear and variance based, so it can miss nonlinear effects and time dependence, and it can be sensitive to outliers if the data include unusual days. Ethically, the main risk is overclaiming. The write-up should avoid using the results to justify decisions that affect people without additional checks, and it should be clear about uncertainty, context, and what the dataset does not measure.

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A AI PROMPTS

1. Check my sample covariance derivation and tell me if any step is wrong.
2. Check if my centered data covariance setup is written correctly and explain the n minus one part in simple words.
3. Is my explanation of why n minus one is used for the unbiased estimate ok.
4. Tell me that PCA directions come from the covariance matrix eigenvectors..
5. Check that the first principal component being the max variance direction is correct.
6. Ss my percent variance explained calculation from eigenvalues is correct.
7. PCA scores are uncorrelated for the figure?
8. Check that my factorization theorem argument for sufficiency is correct.
9. Confirm that for a normal model with unknown mean and variance the sum of samples and sum of squared samples is enough as a sufficient statistic.
10. Check my Cramer Rao lower bound setup for estimating the mean.
11. Confirm whether the sample mean is efficient for estimating the mean under normal assumptions.
12. Is my explanation of what the empirical CDF plot is showing when compared to a reference CDF ok or do i need to change it.
13. Can u check that a negative loading sign in PCA is not an error and the sign can flip.
14. Can you give me a list of human variables effected by this study?